



Matteo Barbetti

Data Scientist

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Summary

- 9+ years of Python development**, applied to scientific computing and industrial solutions.
- 6+ years of hands-on ML experience**, designing and deploying predictive models.
- Former academic researcher in Particle Physics**, with strong foundations in data analysis and simulation.
- AI consultant**, building data-driven solutions for real-world industrial problems.
- Strong interest in emerging technologies**, particularly AI, GenAI, and Quantum Computing.

Strengths

- Analytical problem solver.** Structured, data-driven approach to complex challenges.
- Autonomous, execution-driven.** Owns tasks end-to-end and delivers effective solutions.
- Collaborative, client-focused.** Strong teamwork and clear communication aligned with client needs.

Experience

Data Scientist

Accenture

2024 – Present Modena (IT)

AI consulting role in the Utilities, Resources & Chemicals sector, developing data-driven solutions to optimize industrial systems and decision-making processes:

- Led the design and industrialization of **data-driven optimizers for oil and gas applications**, integrating AI-based surrogate models to calibrate and optimize production systems using field data
- Led multidisciplinary teams of Data Scientists and Software Engineers to develop and deploy **AI solutions within custom web applications** for industrial clients
- Explored the **scalability of industrial optimization problems** on HPC infrastructures and emerging Quantum Computing paradigms
- Delivered **technical training on Anomaly Detection** to client Data Science teams in the materials industry
- Contributed to the design of **AI and GenAI proposals** for client tenders

Technological Researcher

INFN-CNAF

2023 – 2024 Bologna (IT)

Research position within the ICSC project, contributing to the operation of large-scale cloud and data center services:

- Extended the LHCb fast simulation framework to run on **federated and heterogeneous computing infrastructures**
- Served as the data center technical contact for LHCb, monitoring the use of **large-scale storage and computing resources** for simulation workflows
- Designed and implemented a **GenAI-powered chatbot** to enhance user support services at the data center

Doctoral Student

University of Firenze

2020 – 2024 Firenze (IT)

Ph.D. at the intersection of Particle Physics and Computer Science, applying AI and scalable computing to scientific simulation:

- Designed a cloud-based microservice orchestrating **large-scale optimization workflows** across on-premise, cloud, and HPC resources
- Developed and optimized **GAN-based models for fast simulation** of the LHCb detector response using heterogeneous infrastructures
- Integrated the optimized models into a pure Python simulation framework, achieving up to $\times 1000$ speed-up and **accelerating Physics analyses**

Research Intern

CERN

2019 Geneva (CH)

Selected for a three-month research internship at CERN to explore the use of **Deep Generative Models** (in particular GANs) for **fast simulation** within the LHCb experiment, marking my first hands-on experience with AI-driven approaches in large-scale scientific computing

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Technical Skills

Data Science Toolkit

- > Data handling:
numpy, pandas, geopandas
- > Data visualization:
matplotlib, seaborn
- > Classical ML models:
scikit-learn, xgboost
- > Neural networks:
pytorch, tensorflow, keras
- > Model optimization:
optuna, pyomo, gurobipy

Scalable Computing

- > Cloud Computing: Azure & AWS
(compute, storage, ML services)
- > High-Performance Computing:
SLURM-based batch systems
- > GPU-powered acceleration:
CUDA-enabled workflows

MLOps & Deployment

- > Containerization:
Docker, Singularity / Apptainer
- > Orchestration: Kubernetes
- > CI/CD pipelines:
GitHub Actions, GitLab CI

Memberships

Italian National Research Center
for HPC, Big Data and Quantum
Computing (ICSC)

2023 – 2024 Research

LHCb Collaboration at CERN

2017 – 2024 Research

Italian National Institute for Nu-
clear Physics (INFN)

2017 – 2024 Research

Italian Association of Physics Stu-
dents (AISF)

2017 – 2021 Community

Languages

Italian ● ● ● ● ●
English ● ● ● ● ●

Education

Ph.D. in Smart Computing

University of Firenze

2020 – 2024 Firenze (IT) Score: Ph.D. entitled *cum laude*

Thesis: Development of a scalable fast simulation framework for the LHCb experiment at CERN, based on Deep Generative Models and large-scale training on HPC and Cloud systems

M.Sc. in Particle Physics

University of Firenze

2017 – 2020 Firenze (IT) Score: 110/110 *cum laude*

Thesis: Design of a Python-based simulation framework using AI surrogate models for fast detector simulation in the LHCb experiment at CERN

B.Sc. in Physics and Astrophysics

University of Firenze

2013 – 2017 Firenze (IT) Score: 110/110

Thesis: Data analysis and validation of simulated Particle Physics datasets from the LHCb experiment at CERN

Licenses & Certifications

Reinvention with Agentic AI

Issued by: Accenture Badge 2026

Integrating Quantum and High-Performance Computing

Issued by: IBM Badge 2026

Quantum Machine Learning

Issued by: IBM Badge 2026

AWS Knowledge: Amazon Braket - Training Badge

Issued by: Amazon Web Services Badge 2026

Outreach & Dissemination

Plenary speaker

2024 Fondazione Uniser Pistoia

Invited plenary talk on Artificial Intelligence applications in Physics, delivered within a science outreach seminar series organized by Uniser Pistoia (Tuscany, Italy)

Plenary speaker

2023 Festival della Scienza Alto Vicentino

Invited plenary talk on Artificial Intelligence at a city-wide science outreach festival held in Schio (Veneto, Italy)

Book author

2021 Sassi Junior

Contributor, on behalf of INFN, to a children's science book featuring a section dedicated to Artificial Intelligence

July 14, 2026

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